

you don't have

WEEKENDS.

a talk about agent runtimes

Alagu Praklaya | MLE @ ASAPP | Docker Captain

*"apologies was away all weekend —
catching up now."*

— ops-agent

Every one of us has built this bug.

the loop.

CONTEXT

everything the model can see on this call. tokens in a window. not persistent.

LOOP

model reasons → picks tool → tool runs → result feeds back → repeat.

HARNESS

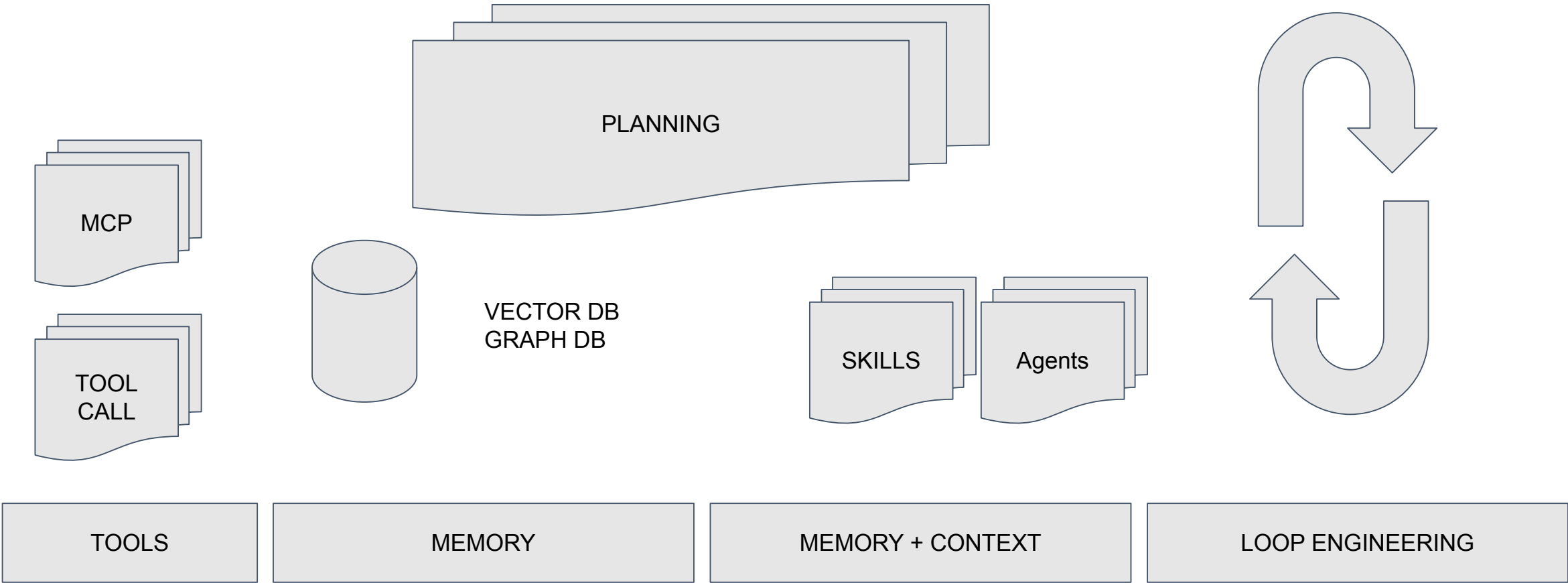
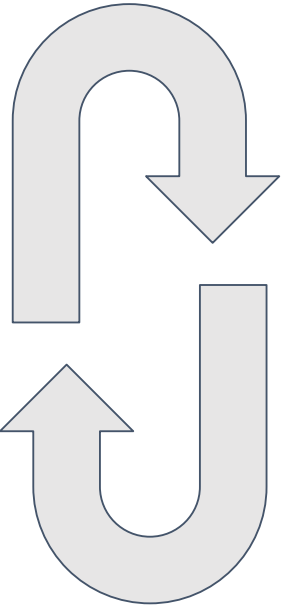
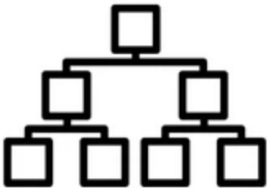
the code around the loop. prompts, retries, safety, memory, orchestration.

you can swap models, swap tools, even swap protocols. change the harness and you have a new agent.

EVOLUTION OF AI HARNESS



LLM



two speeds of thought.

SYSTEM 2

slow. deliberate. costly.

Claude · GPT · Gemini

1 - 5 seconds

does the reasoning.

SYSTEM 1

fast. reflexive. cheap.

Jev · phi3:mini · Haiku

50 - 500 ms

makes the small decisions.

the gateway lives in System 1. the agent lives in System 2. that is the whole architecture.

the protocol soup.

MCP

Model Context Protocol

how tools speak

Anthropic · 2024 · spec revised July 2026

OpenGAP

Open Git Agent Protocol

how identities travel

Lyzr · 2026 · agents as git repos

A2A

Agent-to-Agent Protocol

how agents delegate

Google · 2025 · not widely adopted yet

three protocols, three concerns. none of them tell you where things run.



MEMORY

IDENTITY

CONTINUITY

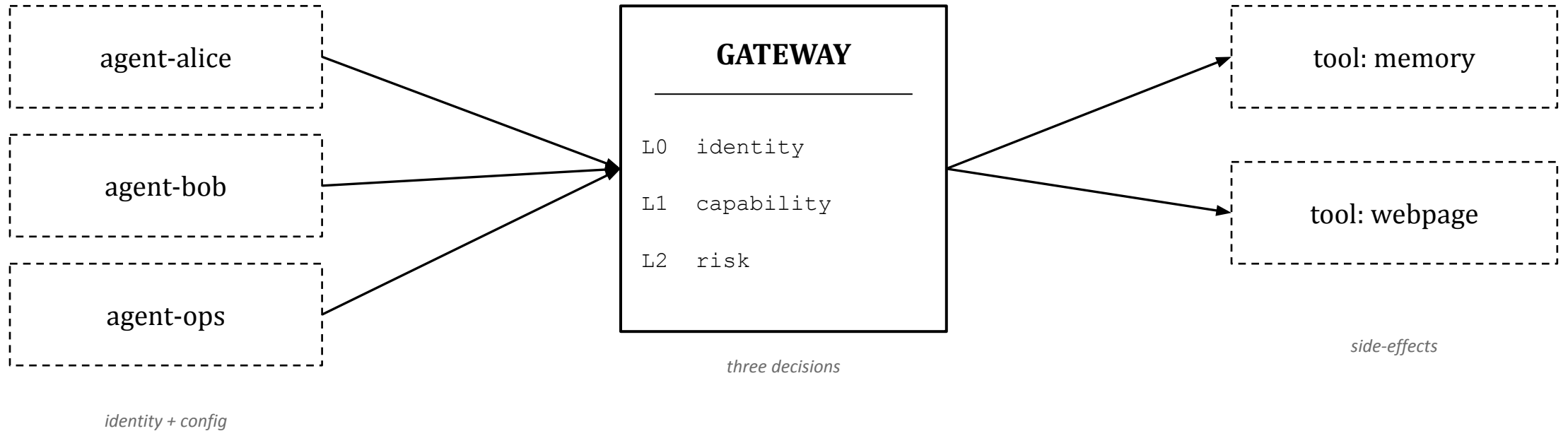
BOUNDARIES

these are

runtime problems.

not model problems.

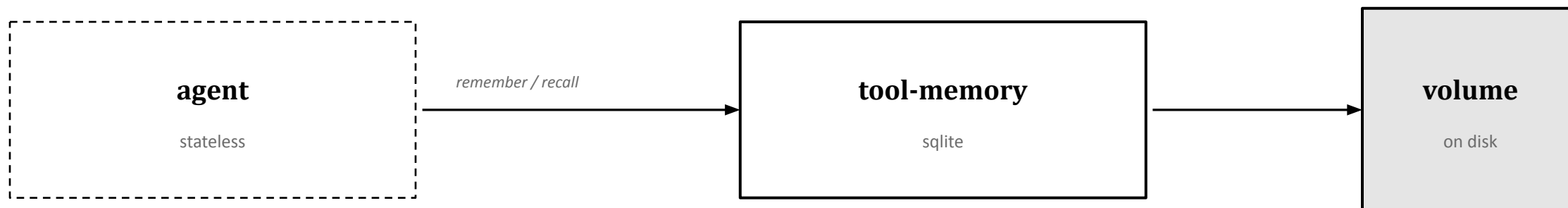
agents are a container problem in disguise.



no new framework. every piece is already in your Dockerfile.

MEMORY

the agent process has none. an external store does.

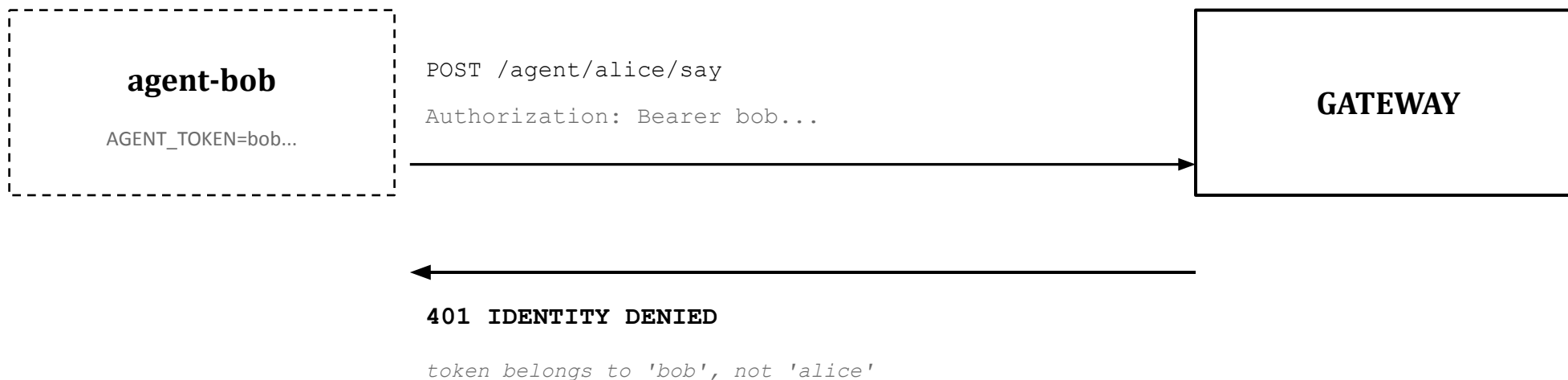


what the runtime provides:

a named volume the model does not know exists.

IDENTITY

who is asking? the URL claims. the token proves.

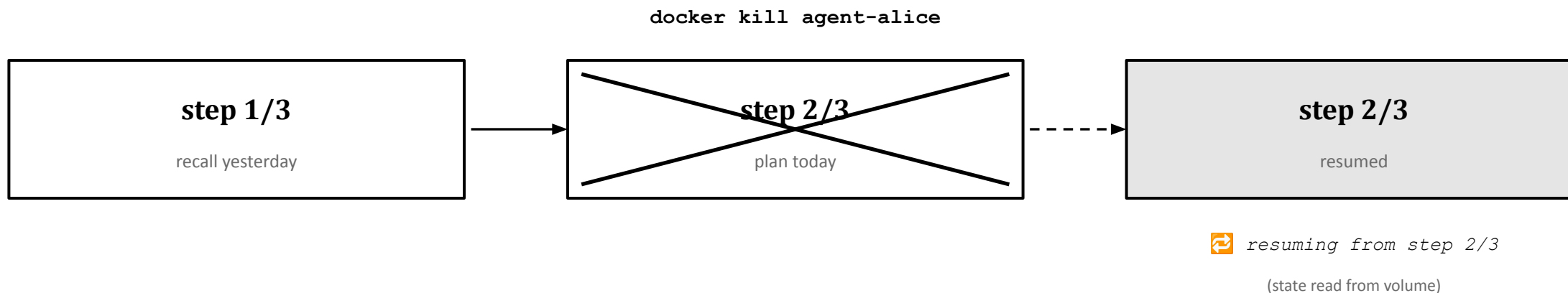


what the runtime provides:

a bearer token per container, checked on every call.

CONTINUITY

processes die. state on a volume does not.



what the runtime provides:

task state on disk. every step is a checkpoint.

BOUNDARIES

what if the model is wrong? every call goes through three doors.



what the runtime provides:

three refusals between the model and the world.

what I did not use.

~~*langchain*~~ ~~*langgraph*~~ ~~*crewai*~~

~~*autogen*~~ ~~*pydantic-ai*~~ ~~*openai agents sdk*~~

not because they're wrong.

because the four things I wanted to show

live one abstraction lower.

STANDING ON

what I did use.

OpenGAP

agents as git repos

lyzr

Ollama

System 1 in a sidecar

self-hosted, phi3-mini

Docker

the runtime, always was

MCP

how tools speak

spec July 2026

FastAPI

the gateway is boring

under 300 lines

Claude

the model, doing what it's good at

if you remember one thing:

**treat MCP servers
like microservices.**

*treat agent runs
like ephemeral jobs.*

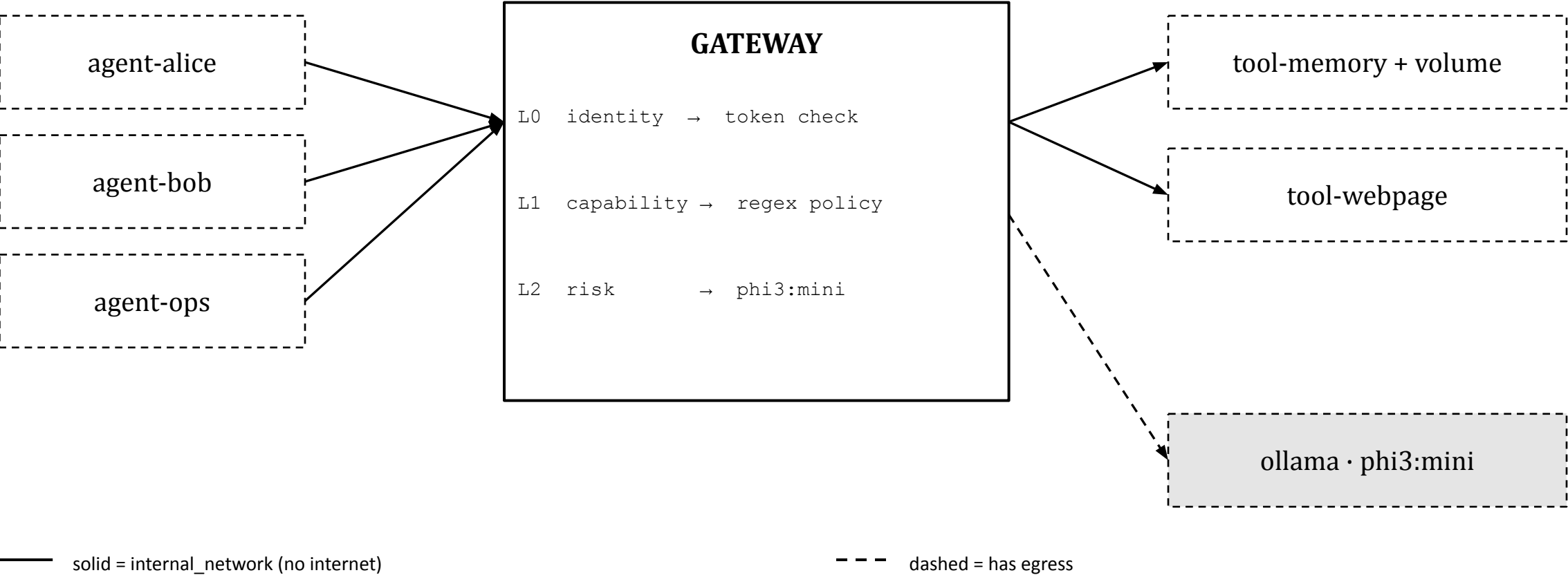
you now have

one agent with a self.

what happens when twenty of them
need to ship a feature together?

that's the next talk.

the full picture.



IF THIS WAS USEFUL

keep going.

REPOS

<code>open-gitagent/gitagent</code>	<i>OpenGAP reference impl</i>
<code>ollama/ollama</code>	<i>the model runtime</i>
<code>modelcontextprotocol/servers</code>	<i>reference MCP servers</i>
<code>docker/mcp-gateway</code>	<i>prior art for what I built</i>
<code>anthropics/claude-code</code>	<i>harness engineering, in production</i>
<code>punkpeye/awesome-mcp-servers</code>	<i>when you need an example</i>

READS

MCP Dev Summit NA 2026 talks	<i>start with 'Sessions Are Dead'</i>
Anthropic — 2026 Agentic Coding	<i>state-of-the-field report</i>
Lyzr — Managing Agents with OpenGAP	<i>the identity-vs-execution split</i>
Simon Willison's weblog	<i>weekly tracker of the field</i>
Latent Space — MCP episodes	<i>podcast, ~2 hours well spent</i>
kernel.org · cgroups & namespaces	<i>how containers actually work</i>

nothing here is required. all of it is worth an evening.

FURTHER READING

you don't have weekends



smarter models.
longer chains of thought.
bigger context windows.



a volume the model doesn't know about.
three no's at the gateway.
a container that came back from the dead.

agent-runtime-demo

*clone tonight.
runs on your laptop.
phi3:mini pulls itself.*

Alagu Prakalya

MLE · Docker Captain

exploring: where MLE meets platform. talk to me at lunch.

